

TrES-3b

R-0077

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_250411 · created 2026-07-15T20:13:31+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460776.9243 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.051
Transit depth (Rp/Rs)^2	3.47 +/- 1.9 [%]
Orbital Inclination (inc)	82.81 +/- 1.86
Airmass coefficient 1 (a1)	0.8503 +/- 0.0083
Airmass coefficient 2 (a2)	0.16 +/- 0.18
Scatter in the residuals of the lightcurve fit is	0.97 %
Best Comparison Star	#2 - [293, 339]
Optimal Method	PSF photometry
Transit Duration (day)	0.0295 +/- 0.0087

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.186 $\pm$ 0.051 vs 0.16309 (deviation 0.45 σ)
Duration fit vs published	0.0295 d vs 0.0567 d (deviation 3.12 σ)
Mid-time O-C	-4.9 min
Depth SNR	3.6
Residual scatter	0.97 %

Light curve

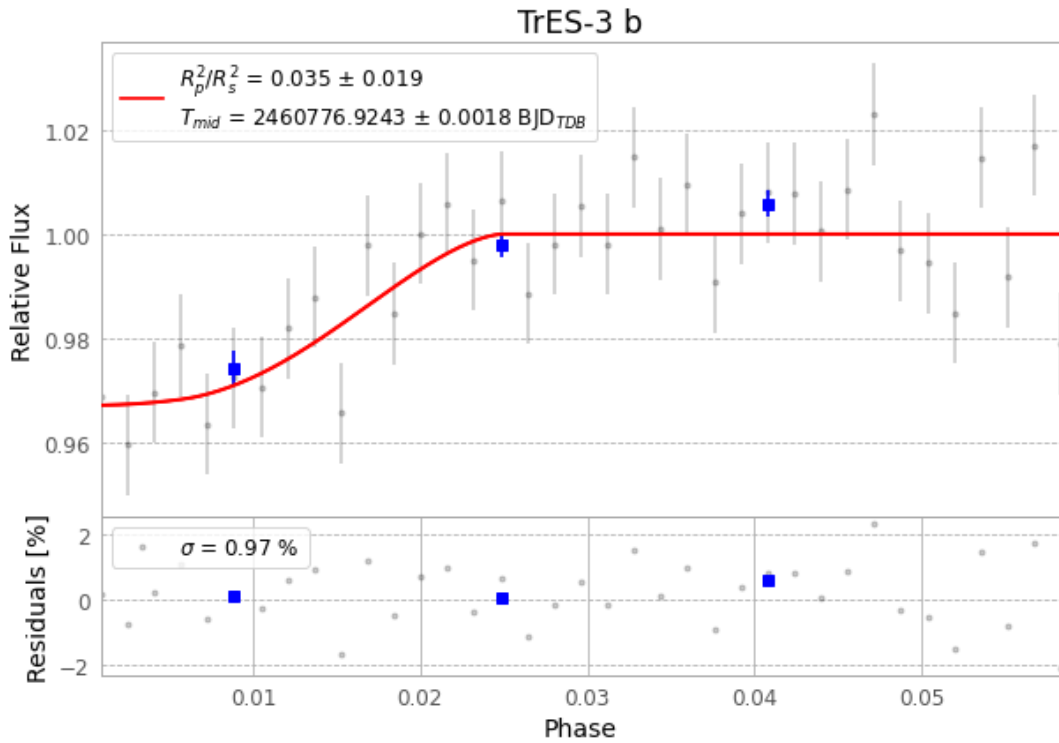


Figure 1 — FinalLightCurve_TrES-3 b_2025-04-11.png (SHA-256 b95fc5b49ead5568...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 09af1deaa90a7156...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

37 FITS frames (24 MB), TRES-3250411100916.FITS → TRES-3250411115715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250411.log (43d38abf3df7...), FinalLightCurve_TrES-3 b_2025-04-11.png (b95fc5b49ead...), FinalLightCurve_TrES-3 b_2025-04-11.csv (7dbe3ca1b46f...)

Compute resources

Wall time 180.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 20:13Z.